

EE133/233 FM Radio Receiver

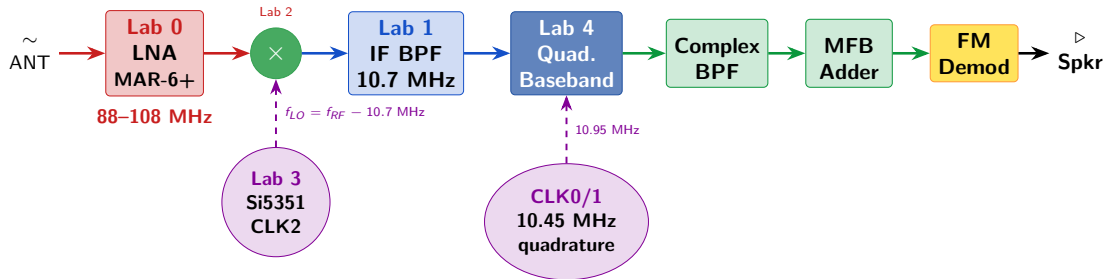
System Overview: Final Project

Edison Altamirano (006948159) Sahan (006948160)

EE133/233: Stanford University

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Full System Architecture



RF Stage

Antenna $\rightarrow$ LNA $\rightarrow$ Mixer
88–108 MHz $\rightarrow$ 10.7 MHz

IF Stage

IF BPF selects 10.7 MHz
Image rejection + channel
filtering

Baseband

I/Q $\rightarrow$ Complex BPF $\rightarrow$ FM
demod
250 kHz $\rightarrow$ audio

Lab 0: MAR-6+ Low-Noise Amplifier

What it does

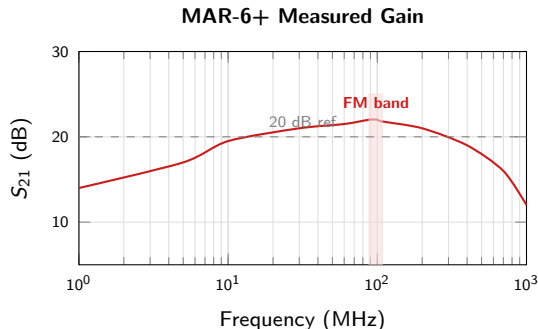
The MAR-6+ is the first stage amplifier. It sets the noise floor for the entire receiver (Friis equation):

$$F_{\text{sys}} \approx F_1 + \frac{F_2 - 1}{G_1}$$

With $G_1 \approx 22$ dB, noise from all later stages is suppressed by $\times 158$.

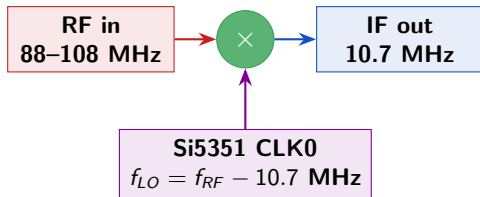
Parameter	Value
IC	MAR-6+ (Mini-Circuits)
Gain	≈ 22 dB @ 100 MHz
NF	2.3 dB
Bandwidth	1 MHz – 1 GHz

S_{21} gain vs frequency (NanoVNA measurement)



Lab 2: ADE-1 Diode Ring Mixer: First Downconversion

Frequency translation

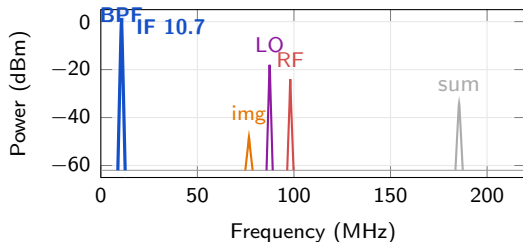


Example (tuned to 98.1 MHz):

$$f_{LO} = 98.1 - 10.7 = \mathbf{87.4 \text{ MHz}}$$

$$f_{IF} = |f_{RF} - f_{LO}| = \mathbf{10.7 \text{ MHz}}$$

Mixer output spectrum (RF = 98.1 MHz)



Product	Formula	Value
IF	$ f_{RF} - f_{LO} $	10.7 MHz
Image	$f_{LO} - 10.7$	76.7 MHz
LO leak	f_{LO}	87.4 MHz
Sum	$f_{RF} + f_{LO}$	185.5 MHz

Lab 3: Si5351 LO Quadrature Generation

Silicon Labs Si5351 Fractional-N PLL

- Programmable PLL controlled via I²C
- Controlled by **Raspberry Pi Pico 2W** over WiFi HTTP
- **CLK2**: first LO for Lab 2 mixer
 $f_{CLK0} = f_{RF} - 10.7 \text{ MHz}$
Covers 77.3–97.3 MHz (full FM band)
- **CLK0/CLK1**: quadrature LO at 10.95 MHz
VCO runs at $64 \times f_{out}$ on PLLA

Quadrature clock generation (Pico 2W firmware):

```
// CLK0 = 0°, CLK1 = 90° (both on PLLA)
si5351.set_freq(quad_hz * RATIO, PLLA, CLK0);
si5351.set_freq(quad_hz * RATIO, PLLA, CLK1);

si5351.set_phase(CLK0, 0); // 0°
si5351.set_phase(CLK1, 64); // 90°
si5351.pll_reset(PLLA); // lock phase
```

Lab 3: Station Tuning and Phase Control

Station tuning via HTTP API:

// First LO: tune to FM station X:

```
GET /api/clk2/set?mhz=X
```

```
CLK2 = X - 10.700 MHz
```

// Quadrature LO: fixed at 10.95 MHz:

```
GET /api/quad/set?mhz=10.95
```

Why phase matters:

- `pll_reset(PLLA)` enforces coherent start
- $\text{RATIO} = 64 \Rightarrow \text{VCO at } 64 \times 10.95 = 700.8 \text{ MHz}$
- $64 \text{ VCO cycles} = \text{one LO period} \Rightarrow \text{exactly } 90^\circ$

FM Interface



Web interface: tunes CLK2 (first LO) and CLK0/1 (quadrature) via WiFi HTTP API

Lab 1: 10.7 MHz IF BPF & Impedance Matching

The problem

Ceramic BPF impedance $Z_0 \approx 330 \Omega$ vs system 50Ω .

Without matching:

- $VSWR = 6.8$ $S_{21} \approx -20$ dB
- Most power reflected; signal lost

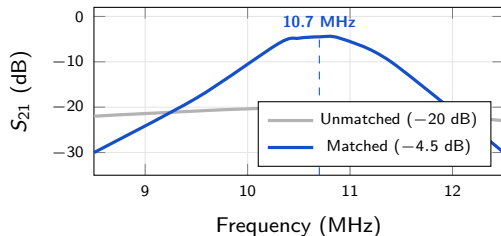
L-network solution ($50 \rightarrow 330 \Omega$):

$$L_s = 1.76 \mu\text{H}, \quad C_p = 106.7 \text{ pF}, \quad Q_T = 2.37$$

With matching:

- $VSWR \approx 1.05$ $S_{21} \approx -4.5$ dB
- **15.5 dB improvement**

S_{21} : matched vs unmatched



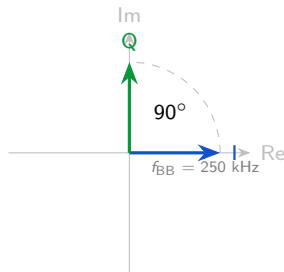
Lab 4: Quadrature I/Q Downconversion

Three cascaded boards:

- 1 **ADT1-4W** transformer: 1:4 impedance, SE $\rightarrow$ differential
- 2 **74CBTLV3253**: quadrature analog switches
Driven at $0^\circ/90^\circ/180^\circ/270^\circ$ by Si5351 CLK1/CLK2
- 3 **AD4940-1** FDA: dual LPF + gain = 2

Note: 74ALVC74 ($\div 4$) not used. Si5351 provides quadrature directly via `set_phase(CLK2, 64)`.

I/Q phasor view



Lab 4: Quadrature I/Q Downconversion

Frequency math:

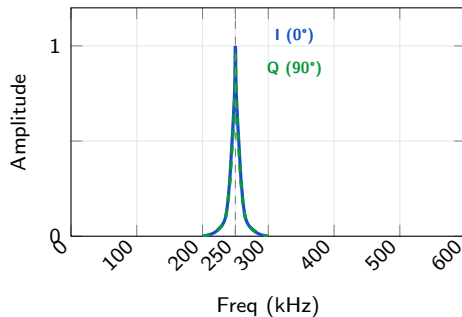
$$f_{\text{quad}} = 10.95 \text{ MHz}$$

$$f_{\text{low-IF}} = |f_{\text{IF}} - f_{\text{quad}}| = |10.70 - 10.95| = \mathbf{250 \text{ kHz}}$$

Measured result:

I and Q outputs both show tone at 250 kHz.

Spectrum at baseband:



Experiment 1: Quadrature + LPF, Hardware Validation

Setup: ADT1-4W + 74CBTLV3253 + AD4940-1 LPF (no complex BPF)

$$\text{IF} = 10.7 \text{ MHz} \quad f_{\text{quad}} = 10.95 \text{ MHz} \quad \Rightarrow \text{low-IF} = 250 \text{ kHz}$$

Oscilloscope (I/Q waveforms)

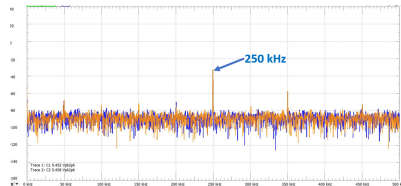


C1 (orange) = I channel C2 (blue) = Q channel

Both at 250 kHz, balanced amplitudes ≈ 5.45

$V_{\text{pk-pk}}$

Spectrogram (Digilent WaveForms)



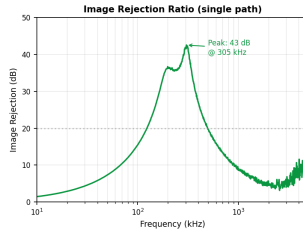
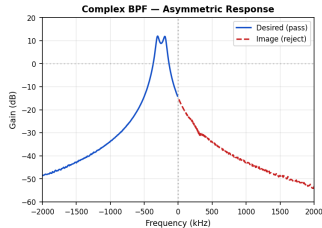
Tone at 250 kHz clearly visible in both channels.

Both I and Q pass the desired tone: image rejection not yet applied.

Complex BPF: Simulation Results

Single-path measurement (polyproc.m, esI/esQ)

Asymmetric response: desired pass vs image reject, + image rejection ratio:



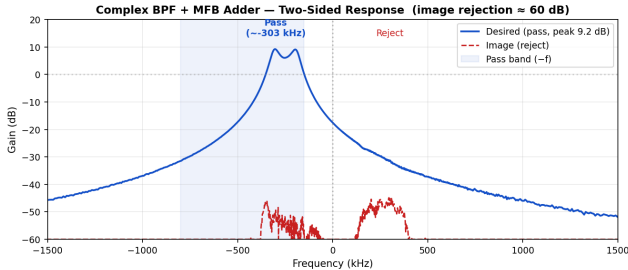
Cross-coupled Tow-Thomas (4× THS4551)

- Cross-coupled I/Q paths $\Rightarrow$ **asymmetric** frequency response
- Passes -300 kHz sideband (desired)
- Rejects $+300$ kHz sideband (image)
- Single path: $\approx$ **43** dB rejection

MFB Adder: Simulation Results

Full chain (ActProcB.m, espl/Q + esml/Q)

Complex BPF + MFB adder two-sided response:



MFB Adder ($1 \times$ THS4551)

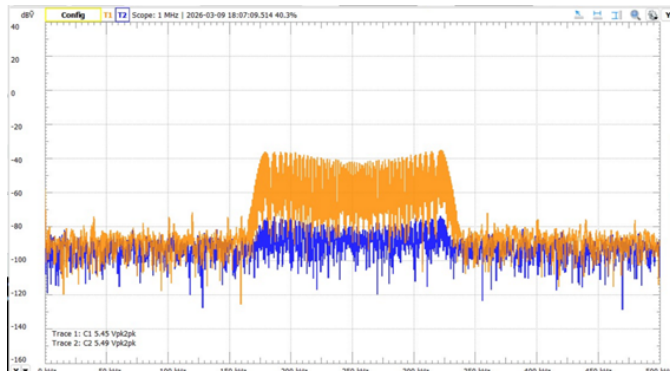
- Combines $I' + Q'$ into one real differential signal
- Full chain rejection: ≈ 60 dB at 303 kHz
- Output feeds FM demodulator

Path	Image Rej.
Single path	≈ 43 dB
Full chain	≈ 60 dB

Experiment 2: Full Chain with FM Demodulation, Hardware Validation

$$\begin{aligned} \text{LO} &= 24.3 \text{ MHz} & \text{RF} &= 35 \text{ MHz} & f_{\text{quad}} &= 10.45 \text{ MHz} \\ f_{IF} &= |35 - 24.3| = 10.7 \text{ MHz} & \Rightarrow & & f_{BB} &= |10.7 - 10.45| = 250 \text{ kHz} \\ \text{FM modulation: } f_m &= 1 \text{ kHz, } \Delta f = 75 \text{ kHz} & \Rightarrow & & \text{Carson BW} &\approx 152 \text{ kHz} \end{aligned}$$

Spectrum after Complex BPF + MFB Adder



Frequency chain:

$$f_{IF} = |f_{RF} - f_{LO}| = 10.7 \text{ MHz}$$

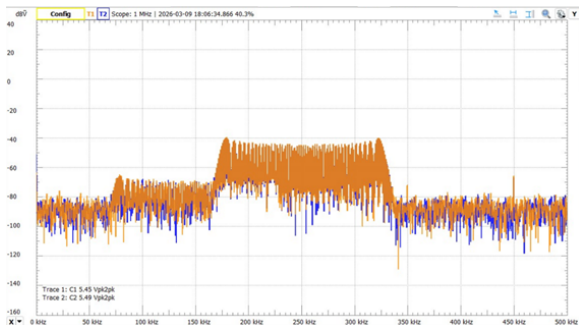
$$f_{BB} = |f_{IF} - f_{\text{quad}}| = 250 \text{ kHz}$$

Parameter	Value
f_{LO}	24.3 MHz
f_{RF}	35.0 MHz
f_{quad}	10.45 MHz
f_{BB}	250 kHz

Performance Metrics: Image Rejection, Before vs After Complex BPF

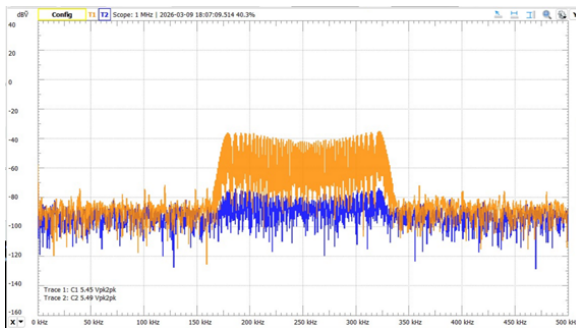
Same setup: LO = 24.3 MHz, RF = 35 MHz, $f_{\text{quad}} = 10.45$ MHz, FM: 1 kHz / 75 kHz dev.

Step 1: Quadrature output only (no Complex BPF)



FM: $f_m = 1$ kHz, $\Delta f = 75$ kHz: all stages up to I/Q output

Step 2: After Complex BPF (250 kHz) + MFB Adder



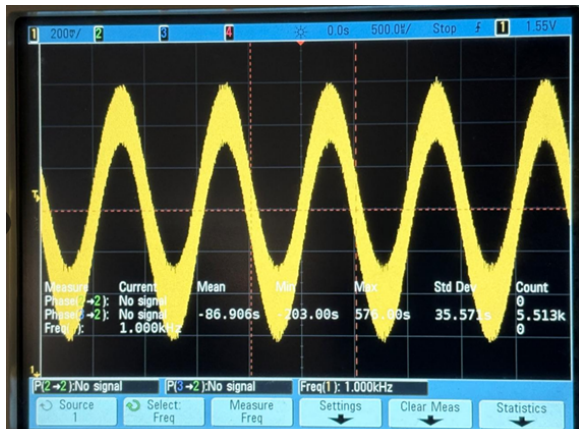
T1 retains FM signal: T2 **significantly reduced and cleaner**

Cross-coupled Taw, Thomas: ≈ 60 dB image

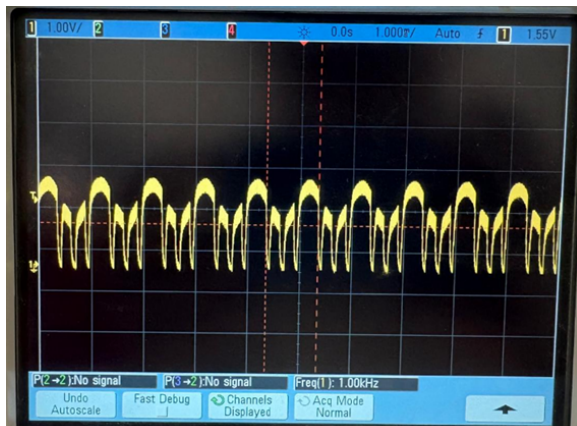
Performance Metrics: Noise Floor, Audio Demodulation (Exp. 3)

LO = 24.3 MHz, RF = 35 MHz, $f_{\text{quad}} = 10.45$ MHz FM: $f_m = 1$ kHz, $\Delta f = 75$ kHz
Varying RF input amplitude

RF = -34 dBm: audio recovers cleanly

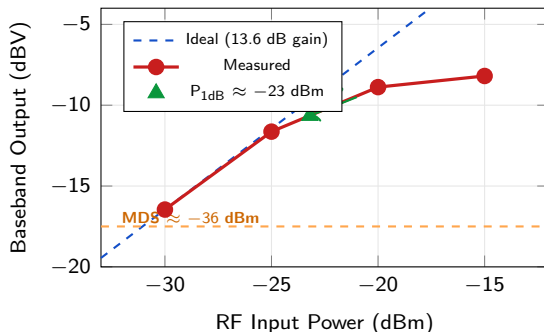


RF = -38 dBm: signal lost / distorted



Performance Metrics: 1 dB Compression Point

Same chain, no FM modulation, sweep RF input, measure baseband output level



Measured data (no modulation):

P_{in}	V_{out}	Gain
-30 dBm	-16.45 dBV	13.6 dB
-25 dBm	-11.64 dBV	13.4 dB
-20 dBm	-8.88 dBV	11.1 dB
-15 dBm	-8.19 dBV	6.8 dB

Key performance summary:

Metric	Value
Small-signal gain	≈ 13.6 dB
P_{1dB} (input)	≈ -23 dBm

FM Demodulation Board

FM Demodulation Board

- Inputs: In+, In−, VDD, GND (differential)
- FM discriminator: Δf deviation $\rightarrow$ voltage
- Audio band recovered: 20 Hz – 15 kHz $\rightarrow$ Speaker

Key measurements:

- **LNA:** +22 dB gain, 2.3 dB NF (FM band, flat)
- **IF BPF:** +15.5 dB after L-network matching
- **Quadrature:** 250 kHz low-IF, balanced I/Q confirmed
- **Complex BPF:** $\approx$ 60 dB image rejection
- **Audio:** recovered at speaker ✓

Conclusion: End-to-End FM Receiver

One station's journey (98.1 MHz):

Stage	In	Out
Lab 0 LNA	-70 dBm RF	-48 dBm RF
Lab 2 Mixer	98.1 + 87.4 MHz	10.7 MHz IF
Lab 1 BPF	mixed products	10.7 MHz clean
Lab 4 I/Q	10.7 MHz IF	250 kHz I/Q
CBF	± 250 kHz	-250 kHz only
Adder	I', Q'	combined real
Demod	FM signal	audio

Antenna → **Speaker**

88–108 MHz → 10.7 MHz → 250 kHz →
audio